Aortic valve replacement in non-elderly: the gap between reality, guidelines and evidence

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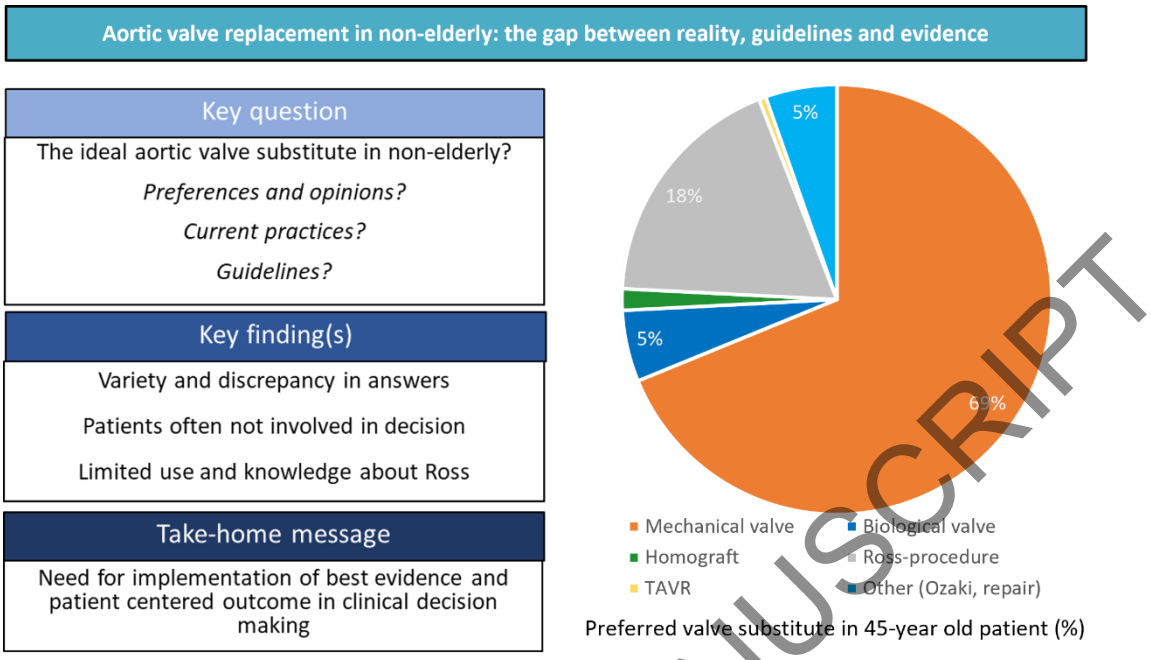
VISUAL ABSTRACT



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ABSTRACT

Objective: There are several treatment options for non-elderly adults (18-60 years) in need of aortic valve replacement (AVR): a mechanical or biological substitute, a homograft, the Ross procedure, or nowadays even transcatheter aortic valve replacement (TAVR). This study evaluated the current opinions and practices of cardiac surgeons and cardiologists concerning aortic valve replacement strategies in this patient group.
Methods: A 43-item online survey was sent out to several scientific organisations, individual cardiac surgeons, cardiologists, and residents/fellows worldwide.
Results: Two-hundred-twenty-two physicians from 33 different countries answered the survey. The United Kingdom (24%), Belgium (23%), and North America (22%) are best represented. A mechanical valve is the most frequently used substitute in non-elderly. With increasing patient age, the popularity of a biological valve increases. TAVR and the Ross procedure are each used in less than 10% of this cohort. Only 12% of the cardiac surgeons has experience with the Ross surgery. In 23% the patient seems not to be involved in the decision on the type of valve substitute.
Conclusion: The preferred surgical treatment for non-elderly patients in need for aortic valve replacement differs widely. To enable a shared decision-making process including patient preferences, there is more need for collaboration between cardiac departments, as well as education and scientific research about patient centred outcomes of the different AVR techniques.

Keywords: young adults, aortic valve replacement (AVR), Ross procedure, transcatheter aortic valve replacement (TAVR), guidelines

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INTRODUCTION

The surgical repertoire for aortic valve replacement (AVR) is diverse, consisting of mechanical valve replacement, biological valve replacement, homograft, and the Ross procedure. In the last decade transcatheter aortic valve replacement (TAVR) has come at lightning speed. Nevertheless, determining the ideal aortic valve substitute for a non-elderly, aged 18-60 years, is challenging due to the heterogeneity in aetiology of aortic valve diseases, potential associated congenital heart defects, associated diseases, active lifestyle, age and expected lifespan. Besides patient characteristics and preferences, local factors like surgical expertise, centre volume, and availability of homografts should be taken into account. Important advantages and disadvantages of the AVR techniques are described in Table 1.

This paper aims to evaluate the current practice and opinions with regards to AVR techniques in non-elderly patients (i.e. less than 60 years old), and whether these findings are in line with the current guidelines: the 2021 ESC/EACTS guideline for the management of valvular heart disease [1] and the 2020 AHA/ACC guideline for management of patients with valvular heart disease [2].

METHODS

Ethics statement

The Research Ethics Committee UZ/KU Leuven approved this research project, (approval number: MP014656, 2020-05-19). Formal written consent from the participants was obtained.

Survey

A web-based tool (Redcap ®, Vanderbilt University, USA) was used to generate a 43-item questionnaire subdivided into six sections: respondent characteristics, centre characteristics, the Ross procedure, applied techniques, cases and advantages of the different AVR techniques (appendix A). The survey was developed after repeated consultation with different specialists.

The survey was distributed via email to 36 different cardiological and cardiosurgical associations (e.g., Belgian Association for Cardio-Thoracic Surgery, Belgian Society of Cardiology, European Society of Cardiology, British Cardiovascular Society, Society for Cardiothoracic Surgery in Great Britain and Ireland, American College of Cardiology, Heart Valve Society). These associations were asked to distribute the survey among their members. The survey was also shared online via Twitter and LinkedIn. The survey was online during 3 months from June 18th 2020 till September 18th 2020.

The collected data was processed anonymously. Numbers of cardiac surgeons per country were obtained from the registry of the Cardiothoracic Surgery Network (CTSNet, http://www.ctsnet.org) in April 2023.

Statistics

Statistical analysis was performed using SPSS (IBM SPSS statistics, Sun Microsystems, US), and Microsoft Excel (Microsoft Office, US). Categorical data were

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expressed as numbers and percentages. Continuous variables were expressed as mean ± standard deviation if normally distributed or median [quartiles] if non-normally distributed. Normality was assessed by the Shapiro-Wilk statistic.

RESULTS

Respondent characteristics (questions 1-10)

Two-hundred-twenty-two physicians responded to the survey. Table 2 shows the characteristics and country of practice of the respondents. The United Kingdom (24%), Belgium (23%), and North America (22%) are best represented. Most participants work in a university hospital (70%), are cardiac surgeons (87%) and male (85%). Seventeen percent of the cardiac surgeons are also congenital cardiac surgeon, with a mean of 129 ± 83 congenital cases each year.

Centre characteristics (questions 11-19)

Figure 2 shows the number of centres in relation to their annual case load, in the best represented countries. The median number of cardiac surgeons per centre is 6.0 [1-50]. In Belgium this is 3.0 [2-10], in the USA 5.5 [1-14], in Canada 6.0 [4-12], in the UK this is 7.0 [3-17], and in Germany 12.0 [4-50].

Figure 3 shows the yearly volume of AVR and TAVR per centre. At least 100 surgical AVR (SAVR) and TAVR interventions are annually performed in respectively 64% and 43% of the centres. In most centres (64%) TAVR procedures are performed by a cardiac surgeon and cardiologist together.

The Ross procedure (questions 20-26)

The Ross procedure is performed in 45% of the centres. Only about 12% of all participating cardiac surgeons believe to be capable to perform the Ross procedure. On the contrary, reasons to 'not' or 'no longer' perform the Ross procedure are: it converts a single-valve disease into a double-valve disease (36%), lack of education (24%), difficult technique (23%), high risk of re-operation (17%), not mentioned in the guidelines or no scientific evidence (14%) and higher morbidity and mortality rates (10%).

Table 3 shows the relation between the surgical AVR volume per centre and the Ross procedure. In the Ross performing centres, the median annual Ross number is 10.0 [1-100] with 69% of these centres performing less than or equal to ten cases per year. The median number of Ross performing surgeons in these specialised centres is 2.0 [1-6]. A mean of 7.3 Ross procedures are performed each year per cardiac surgeon in these centres.

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Applied techniques (questions 27-31)

Table 3 shows also the preferred treatment in a 45-year old patient in relation to different centre sizes. An evaluation of who takes the decision on the type of AVR shows that in 55% this is the Heart Team and in 6% this is autonomously the patient. In contrast, in 23% the patient is not involved in this decision.

In non-elderly, a mechanical substitute is the preferred strategy used in more than 50%. Biological valve replacement is less common with nearly 70% of the respondents offering a biological valve to less than half of the patients. TAVR or the Ross procedure is used in less than 10%.

Case-based decision-making (questions 32-37)

To further analyse the decision-making process of the specialists, five cases with different age or gender are presented, all with a low surgical risk. The answers on the preferred valve substitute, also comparing American and European centres, can be found in figure 4. Mechanical AVR (61%) is most frequently chosen as an option for non-elderly women without a pregnancy wish. If there is a pregnancy wish, the biological valve (46%) and the Ross procedure (39%) are most frequently chosen. For a 45- and 55-year-old patient with a low operative risk, the preferred AVR option is a mechanical valve (respectively 69 and 52%).

Finally, each participant was asked which AVR technique he or she would prefer for themselves taking age into account, see figure 5. The mechanical valve is preferred until the age of 50 years. The biological valve is the most popular option for physicians over 50 years old. No single doctor would choose a homograft for him or herself. The Ross procedure is selected in all age categories up 60 years but is always less popular than mechanical and biological valves.

Advantages of different aortic valve replacement techniques (question 38-43)

The last part of the questionnaire deals with the opinion of the respondents on the different AVR techniques.

<u>Advantages of mechanical AVR in non-elderly:</u>
Most given reasons to choose a mechanical substitute is that it is believed to be a lifelong solution (62%) with a lower rate of re-intervention compared to biological AVR (80%) and the Ross procedure (55%). Moreover, it is associated with a lower surgical risk then the Ross procedure (63%).

<u>Advantages of biological AVR in non-elderly:</u>
The most cited advantage of biological valves is the option of TAVR in case of re-intervention (64%). Furthermore, it is suggested that the new generation biological valves are associated with better durability (62%). The lower operative risk, compared with the Ross procedure, is deemed important (59%).

<u>Advantages of TAVR in non-elderly:</u>
A faster return to daily activities (59%) and shorter hospital stay (59%) are the most chosen advantages of the TAVR technique.

<u>Advantages of the Ross procedure in non-elderly:</u>

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Cited advantages of the Ross procedure are the lower bleeding risk (63%) and the lower risk to develop thromboembolism (75%) compared to mechanical AVR. The third and fourth most chosen benefit is the better quality of life (45%) and better hemodynamics (44%) of the Ross procedure. About one out of four (28%) respondents consider the Ross procedure to be associated with superior survival compared to a mechanical and biological substitute. The most eligible candidates for the Ross procedure are: children (84%), women with a child-bearing age (78%), patients who would otherwise receive a mechanical valve (67%) or athletes (55%). Patients with endocarditis (28%) or in need of a re-operation (27%) are believed not to be ideal to undergo a Ross procedure. On the contrary, 28% of the participants considers endocarditis as a good reason to choose for the Ross procedure.

DISCUSSION

The guidelines

This study, probing into the preferred strategy for aortic valve replacement in non-elderly adults, indicates that there is large heterogeneity in the answers. This heterogeneity is indicative for the lack of strong evidence to support one option over the other. Even most reliable literature is based on matched designs with only few randomised trials [3]. Therefore, preferences should be related to surgical volume and expertise as well as patient's life expectancy, lifestyle, bleeding and thromboembolic risk and the informed patient preference. Therefore it is remarkable that in almost one out of four cases the patient is not involved in the choice of type of AVR, despite Class of Recommendation (COR) 1/I – Level of Evidence (LOE) C recommendation in the guidelines [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**]. Moreover, this questionnaire suggests that not all existing procedural options are presented to each patient. Luckily despite the unclear scientific benefit, majority of patients are discussed within the Heart Team, which is central to both the American and European guidelines, COR 1/I LOE C. These guidelines also agree that high volume centres are associated with better outcomes while acknowledging the influence of other performance indicators [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**]. But neither the American nor European guidelines recommend minimum case load for TAVR and surgical aortic valve replacement. The direct relationship between outcome and volume seems conventional wisdom but this is not proven for standard AVR [4]. As shown in figure 2 and figure 3 there is a huge variation in the annual case load among the respondents. Most participating Belgian centres are small. In contrast, the UK has predominantly medium and large size centres. The USA is represented by some small centres and many large centres. As shown in Table 3 it seems that in smaller units there are less Ross operations performed. This suggests that there might be an association between the size of the centre and the different treatment options offered to an individual patient. It has been shown that for rare and more technical operations, like the Ross operation, volume plays an important role in favouring outcomes. Not only to prevent major complications but also the chance to survive a complication is higher in a high-volume hospital [5].

Cardiologists and cardiac surgeons should have an excellent understanding of the short and long-term outcomes of every aortic valve substitute, considering the current evidence. Looking at the ACC/AHA and ESC/EACTS guidelines, both agree that a

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mechanical prosthesis is recommended in the young, nevertheless they differ in age thresholds [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**]. Patients <50 years (COR 2a LOE B) and <60 years (COR IIa LOE B) should get a mechanical aortic prosthesis in the ACC/AHA, respectively ESC/EACTS guidelines. Nevertheless, as also shown in this inquiry, there is a shift towards implanting biological valves in younger patients. This can potentially be attributed to avoiding anticoagulation in young patients, the assumption that newer generation bioprosthetic valves are long-lasting and that valve-in-valve TAVR can be performed in a degenerated valve. The ACC/AHA (COR 2b LOE B) and ESC/EACTS guidelines (no specific recommendations), respectively, state that a Ross procedure may be considered 'in patients <50 years of age who prefer a bioprosthetic AVR and have appropriate anatomy at a Comprehensive Valve Centre' or 'in appropriately selected patients but high expertise in aortic surgery is required'. They both cite observational studies but consider the evidence insufficient, which can be questioned [6].

Mechanical aortic valve replacement

Thanks to the low reintervention rate, a mechanical substitute is often preferred in young adult patients. Sixty-two percent of the participants in our survey believed a mechanical substitute offers a lifelong solution. Nevertheless, the annual reoperation rate of mechanical AVR is about 0.5% and is mostly associated with urgent indications such as prosthesis failure, endocarditis and thrombosis [7]. Due to the need of oral anticoagulation therapy mechanical aortic valves are associated with a higher risk for thromboembolic and bleeding events, about 1.7% annually [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**]. As stated by several participants, efforts have been made to reduce bleeding events without influencing thromboembolic risk. For example, the Prospective Randomized On-X Valve Anticoagulation Clinical Trial (PROACT) showed that a lower anticoagulation target (in combination with low-dose aspirin) results in a reduced rate of bleeding events, thromboembolic and thrombosis events [8]. Several studies (e.g. PROACT Xa) evaluated novel oral anticoagulants (NOACs), but this does not seem to be an alternative to direct vitamin K antagonists [9]. Moreover, mechanical AVR is associated with a reduced life expectancy when compared to an age matched population. Noteworthy is that in cases younger than 50 years a mechanical valve is only chosen by 37% of the doctors for themselves, while it is proposed to more than 69% of their patients in the same age category.

Biological aortic valve replacement

Bioprosthetic valves do not require anticoagulation therapy and are associated with a significantly lower incidence of stroke and bleeding. This is the reason that in patients with a pregnancy wish it is the preferred valve substitute (46%). Yet, biological valves are less often used in younger patients due to their limited durability and higher cumulative risk of reintervention. These disadvantages might explain why the long-term survival after biological AVR is lower compared to gender- and age matched patients [10]. Special attention should be paid to avoid patient prosthesis mismatch as this is clearly associated with a higher rate of structural valve deterioration and impacts survival [11,12]. More recently, the durability of biological valves has potentially increased with the arrival of new generation biological valves that incorporate several preservation technologies [13]. Furthermore, some surgeons advocate the use of the Ozaki procedure in young patients [14]. As these valves or technique still have short clinical follow-up, conclusions about long-term advantages

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should be interpreted cautiously. In our survey this argument seems important, as 62% stated this was a valid reason to choose a biological substitute in young patients. In addition, 64% believed that transcatheter options are available for subsequent reintervention. It has indeed been shown that TAVR after SAVR is associated with better short-term outcomes than redo SAVR [15]. However, in some patients there is a high-risk for coronary obstruction and the long-term durability of a TAVR-in-SAVR remains uncertain. Remarkable is that in the 45-year-old patient case the biological valve was selected by merely 5% of the specialists, while 32% of the doctors under the age of 50 would choose a biological valve for themselves.

Homograft

Quite like bioprosthetic valves, homografts degenerate exponentially over time. Moreover, reoperations are technically challenging due to extensive calcification of the homograft wall [16]. Therefore, in accordance with the responses, homografts are considered suboptimal substitutes for aortic valve replacement in young adults. While there might be additional value in using decellularized homografts, long-term data are lacking [17].

The Ross procedure

The Ross procedure or pulmonary autograft replacement alleviates the need for lifelong anticoagulation. In several recent reports, this translated into a long-term survival equivalent to the age- and gender-matched general population and superior to mechanical and biological AVR [18]. This was not reflected in the guidelines, neither in the answers, as only 28% thought this was an advantage of the Ross procedure. Besides the survival benefit, it has been shown that the Ross procedure is associated with a lower risk for thrombo-embolic events, bleeding complications and provides better quality of life; advantages only 75%, 63% and 45% respectively of the respondents agreed on. Only 44% agreed that the Ross procedure is associated with superior hemodynamics compared to conventional AVR, something which has been proven even at peak exercise [19]. In this survey the majority of respondents (63%) is convinced that the Ross procedure is associated with a higher surgical risk than a mechanical or biological valve substitute. It is indeed a more extensive operation, yet literature shows that when performed by an experienced surgeon in a high-volume centre there is no difference in perioperative mortality compared to prosthetic AVR [20,21]. In this survey only 8% of all centres and 18% of the Ross performing centres, do more than 25 Ross operations on an annual base. Main reasons not to perform this operation are the technical complexity and the absence of specific education. Of all responding surgeons, 34% have performed the Ross procedure, but more than 55% of them did less than 20 procedures. This is probably not enough to overcome the learning curve. Bouhout et al [22] showed that for the Ross procedure there is an improvement in safety and efficiency after approximately 75-100 cases, a number only 12% of the Ross performing surgeons in this survey achieve. This underlines the need for dedicated training programs. More than half of the respondents (55%) say that a Ross operation is associated with a higher re-intervention rate than mechanical AVR. Within literature this is inconclusive [23]. The Ross procedures is offered to one out of four patients in only 10% of the centres. Nevertheless 19% of the respondents below 50 years of age would prefer a Ross operation for themselves.

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Transcatheter aortic valve replacement (TAVR)

The use of TAVR has grown exponentially since its introduction in 2002. More recently TAVR is approved for all patient risk profiles irrespective of age. The Partner 3 trial [24] and the Evolut Low Risk trial [25] showed that TAVR is a suitable alternative in patients with a low surgical risk. Besides the short-term follow-up in these trials most patients enrolled were in their 70s [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**]. Consequently, there remain many unanswered questions in the younger low-risk patients, like the durability of transcatheter heart valves, future coronary access, how to address concomitant conditions, safety and feasibility of future TAVR-in-TAVR or even TAVR-in-TAVR-in-TAVR [26]. This explains why TAVR is not the gold standard in the young low-risk patient. Additionally, many younger patients have a uni- or bicuspid aortic valve, an exclusion criterion in all large RCTs, which may complicate the transcatheter approach. In our inquiry most centres use TAVR in less than 10% of the young patients. The main reason for choosing a TAVR procedure is the shorter hospital stay and faster return to activities of daily living, both options are selected by 59% of the specialists. Nevertheless, in the 65-year-old low-risk case only 3% of respondents would favour TAVR. In contrast, 6% of participating doctors under the age of 50 would choose a TAVR for themselves. In most centres (64%) TAVR is performed by the surgeon and cardiologist together. In 31% TAVR implantations are performed by the interventional cardiologist solely, especially when using a femoral approach.

Study strengths and limitations

In this survey, 222 cardiac surgeons and cardiologists participated from around the world in all age-categories, although not every type of institution, gender, age group and geographic region are equally represented. Especially low- and middle-income countries are underrepresented. While our survey likely reflects real life clinical practice, it is possible that our study provides an overrepresentation of physicians working in tertiary referral centres. There is a continuous evolution in valve prosthesis, techniques and peri-operative care which should be taken into account.

CONCLUSION

The preferred surgical treatment for non-elderly patients in need for aortic valve replacement differs widely among surveyed specialists. To enable a shared decision-making process while involving patient preferences, there is a need for collaboration between cardiac departments, as well as education and scientific research highlighting patient centred outcomes of the different AVR techniques.

KEY POINTS

What is already known?
• Mechanical valves are associated with higher rates of anticoagulation-related bleeding and thromboembolic events which impact quality of life and life expectancy [**Error! Bookmark not defined.**]. • Bioprosthetic valves in the non-elderly patient in need of aortic valve replacement are associated with limited durability and lower than expected

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survival, especially in the presence of patient-prosthesis mismatch [**Error! Bookmark not defined.**].

- The Ross procedure can restore a normal life expectancy and quality of life in young and middle-aged adults [**Error! Bookmark not defined.**].
- TAVR so far has never been systematically tested in young, low-risk patients [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**].

What does our study add?

- The most preferred AVR techniques in young adult patients (less than 60 years old) are: mechanical valves > biological valves > the Ross procedure > TAVR.
- With increasing patient age, the popularity of a biological valve increases while the popularity of the Ross procedure decreases.
- There is a discrepancy between the AVR option that specialists would choose for themselves and the one they offer to their patients.
- Only a limited number of young patients in need of AVR are offered a Ross procedure.
- The Ross procedure is only in 12% of the cardiac surgeon's toolbox.
- 23% of patients are not involved in the decision of the aortic valve substitute.
- Current guidelines do not reflect the published evidence of the Ross procedure and should be reconsidered [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**].

CONFLICT OF INTEREST

Filip Rega is a consultant to Corcym.
Peter Verbrugghe is a consultant to Corcym.
Bart Meuris is consultant to Edwards Lifesciences, Medtronic and Corcym.

There are no other conflicts of interest.

FINANCIAL DISCLOSURE

There was no financial reward for participating in the questionnaire.

DATA AVAILABILTY STATEMENT

The data underlying this article are available in the article. Additional data gained during the survey can be shared on reasonable request to the corresponding author.

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FIGURES LEGENDS

Central figure: An online survey evaluating aortic valve replacement in non-elderly adults.

Figure 1: Geographic location of the respondents and proportion to the number of cardiac surgeons (registered on CTSNet).

Figure 2: Number of cardiac procedures of the best represented countries performed per centre per year.

Figure 3: Proportion of centres and the number of surgical AVR and TAVR performed annually

Figure 4: Preferred AVR options in American versus European centres for adult patients with a low surgical risk (other: Ozaki procedure – Aortic valve repair).

Figure 5: Preferred AVR technique of the respondents if they themselves needed surgery taking their age into account *(other: Ozaki procedure – Aortic valve repair)*.

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TABLES

	Advantages	Disadvantages
Mechanical AVR [27]	+ Excellent structural valve durability	- Lifelong anticoagulation - Risk of thromboembolic events - Risk of prosthetic valve endocarditis - Impact on life expectancy
Biological AVR [**Error! Bookmark not defined.**,**Error! Bookmark not defined.**]	+ Lower thromboembolic risk compared to mechanical AVR + No need for lifelong anticoagulation therapy + Potential better durability with new generation biological valves + Possibility to receive TAVR valve-in-valve	- Limited/unknown durability due to structural valve degeneration - Risk of prosthetic valve endocarditis - Impact on life expectancy
Aortic homograft [28]	+ Excellent hemodynamic profile + Low thromboembolic risk + No need for lifelong anticoagulation therapy + Low risk of prosthetic valve infection + Potential better durability with decellularized homografts	- Limited/unknown durability due to structural valve degeneration - Higher reoperation risk - Higher cost - Limited homograft availability
Ross-procedure [**Error! Bookmark not defined.**,29,30]	+ Excellent hemodynamic profile + Low thromboembolic risk + No need for lifelong anticoagulation therapy + Low risk of prosthetic valve infection + Restores normal life expectancy	- Potentially converts a single-valve disease into a double-valve disease - More complex surgery and higher risk in low volume centres - Higher reoperation risk - Limited homograft availability
TAVR	+ Smaller incision wound + Faster recovery + No need for lifelong anticoagulation therapy + Possibility to receive TAVR-in-TAVR later in life	- Unknown valve durability in the young - Higher pacemaker implantation rate - Higher rate of paravalvular leakage - Potential compromised coronary access

Table 1: advantages and disadvantages of the most used aortic valve replacement techniques.

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Variable	Number	(%)
Total number participants	222	(100%)
Age in years		
< 30y	5	(2%)
30-40 y	64	(29%)
41-50y	57	(26%)
51-60y	57	(26%)
> 60y	39	(18%)
Gender		
Male	188	(85%)
Female	34	(15%)
Specialisation		
Cardiologist	28	(13%)
Cardiologist in training	2	(7%)
Cardiologist with > 20 years' experience	10	(36%)
Cardiac surgeon	194	(87%)
Cardiac surgeon in training	35	(18%)
Cardiac surgeon with > 20 years' experience	66	(34%)
Workplace		
University hospital	155	(70%)
Regional hospital	60	(27%)
Both	7	(3%)

Table 2: Respondent characteristics and participating countries and number (%) of participants per country.

Number surgical AVR /centre /year	Number respondents	Mean number cardiac surgeons / centre	% Ross performing centres	Mean number Ross operations/ Ross centre/year	Number Ross operations/Ross cardiac surgeon/year	Strategy in a 45-year-old patient (Biological – Mechanical – Ross)(%)
< 25	10	3.0	20%	5.0	2.5	20% - 60% - 10%
25-50	18	4.5	22%	9.8	4.3	0% - 89% - 11%
51-100	39	6.0	40%	9.9	5.2	0% - 79% - 16%
101-200	53	6.0	47%	8.6	5.1	8% - 71% - 15%
201-500	69	8.7	60%	18.2	9.1	5% - 55% 28%
> 500	8	14.6	50%	39.3	13.1	20% - 80% - 0%

Table 3: Relation between the magnitude of the centre and the Ross procedure.

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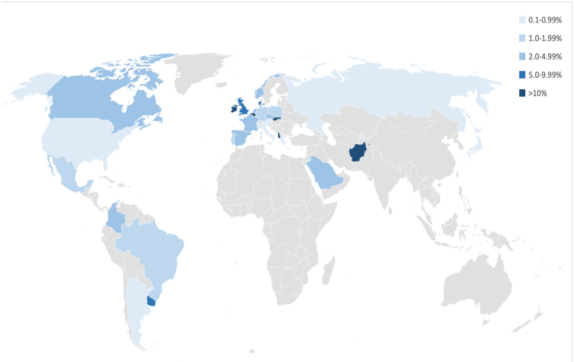
Central image

The ideal aortic valve substitute in the non-elderly patient

Preferences and opinions?

Current practices?

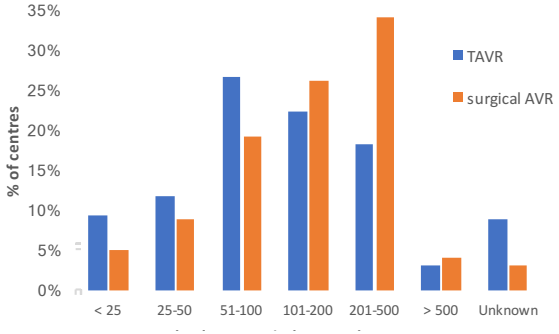
Guidelines?



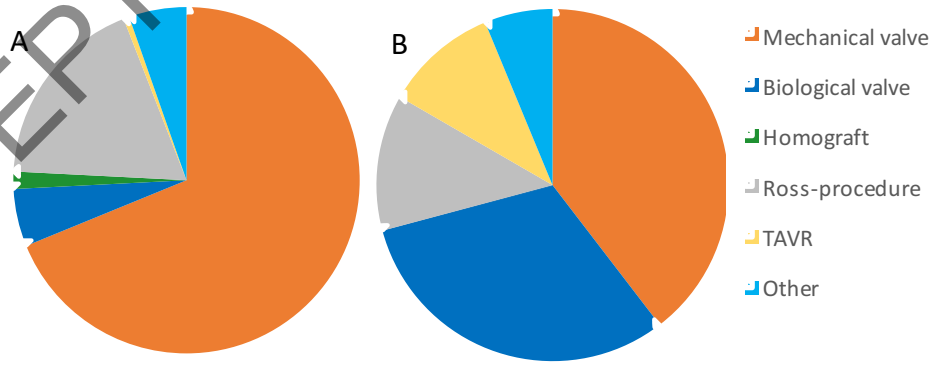
Proportion of respondents to cardiac surgeons/country

Ross?

- Only by 12% of surgeons
- Only 18% of Ross centres > 25 Ross/year
- Offered to < 10% of patients
- Current practice ≠ guidelines ≠ evidence



Annual volume TAVR – surgical AVR per centre



Preferred technique for the patient (45 years (A) and the respondant him/hereself (B) in the age group 40-50 years.

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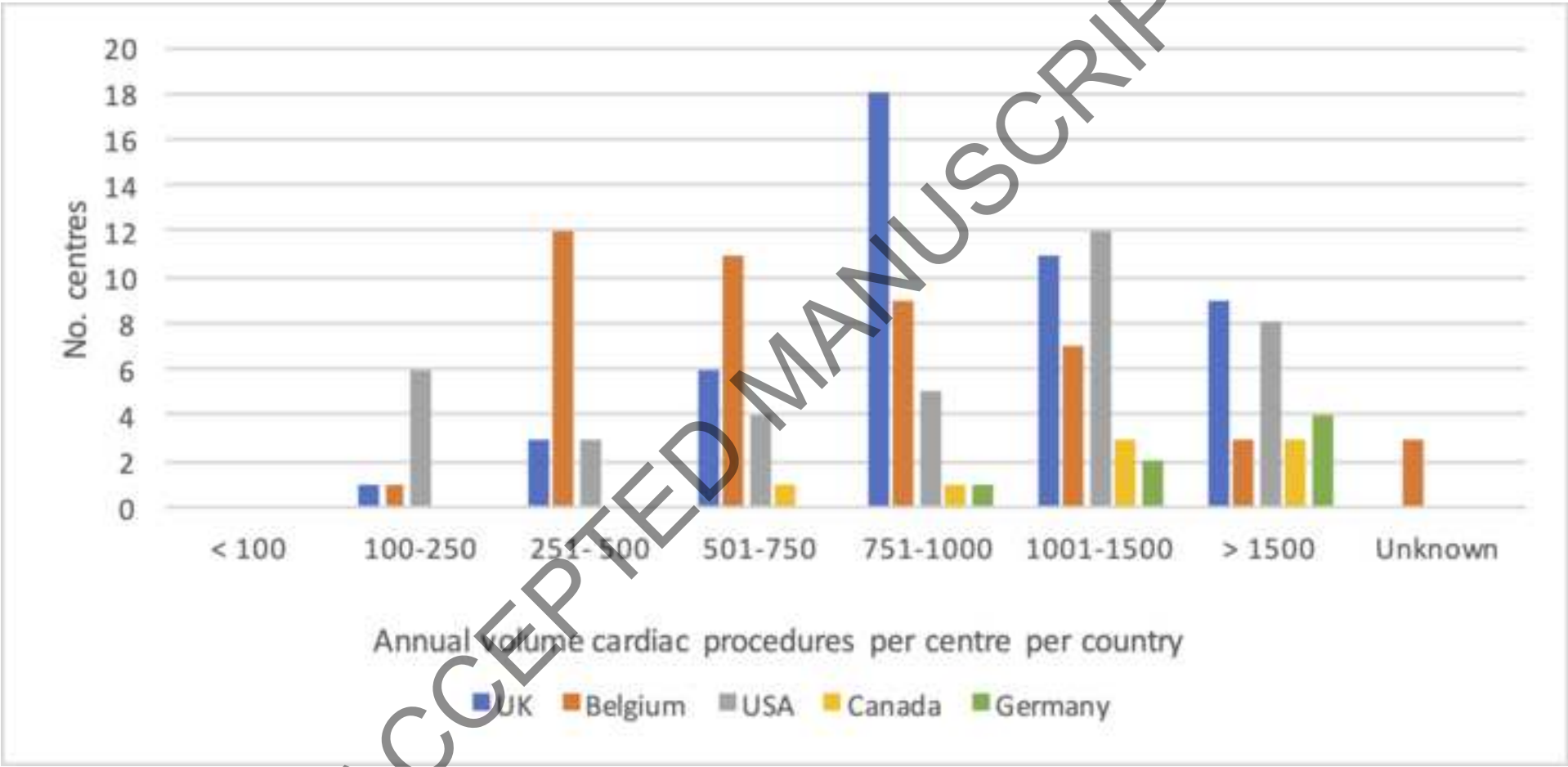

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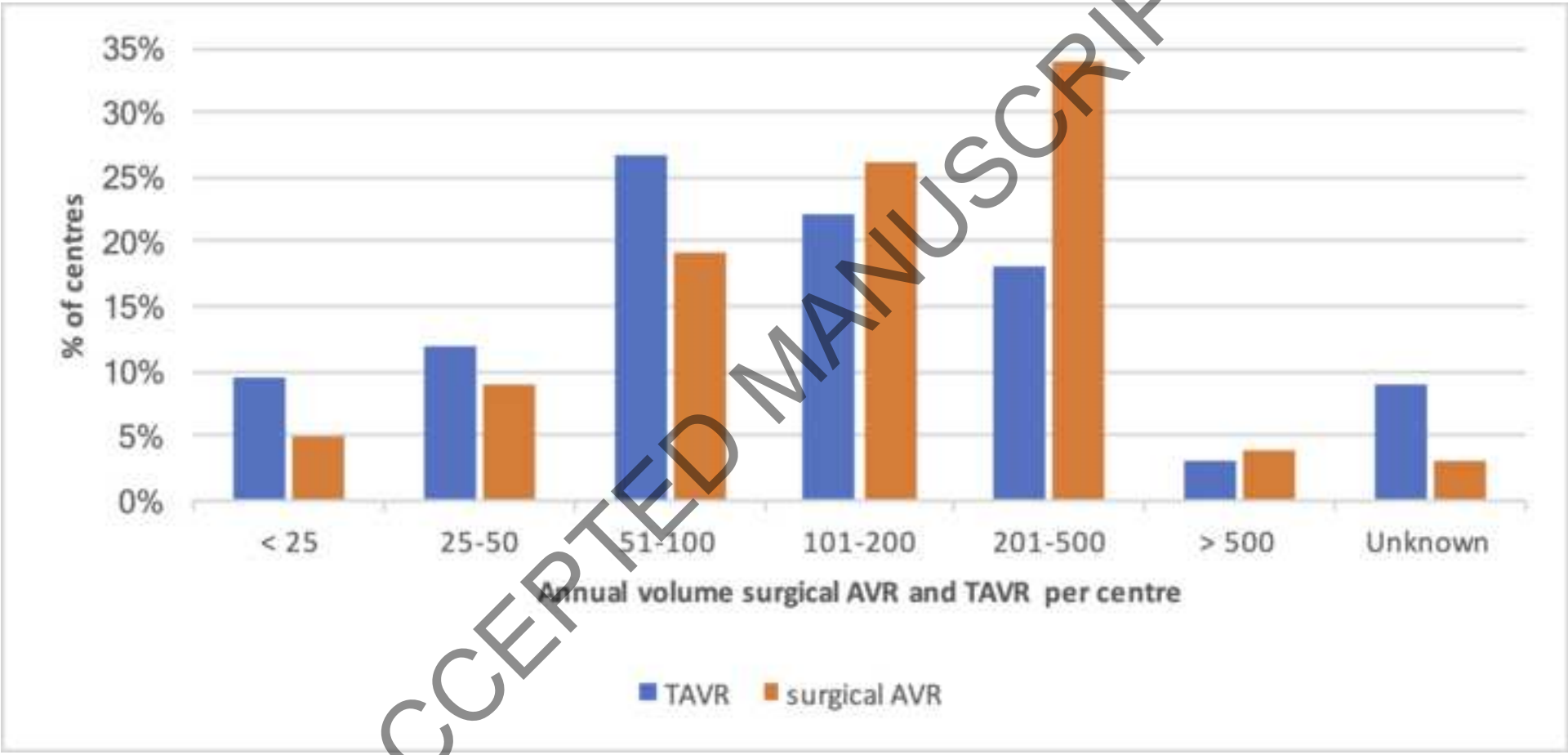

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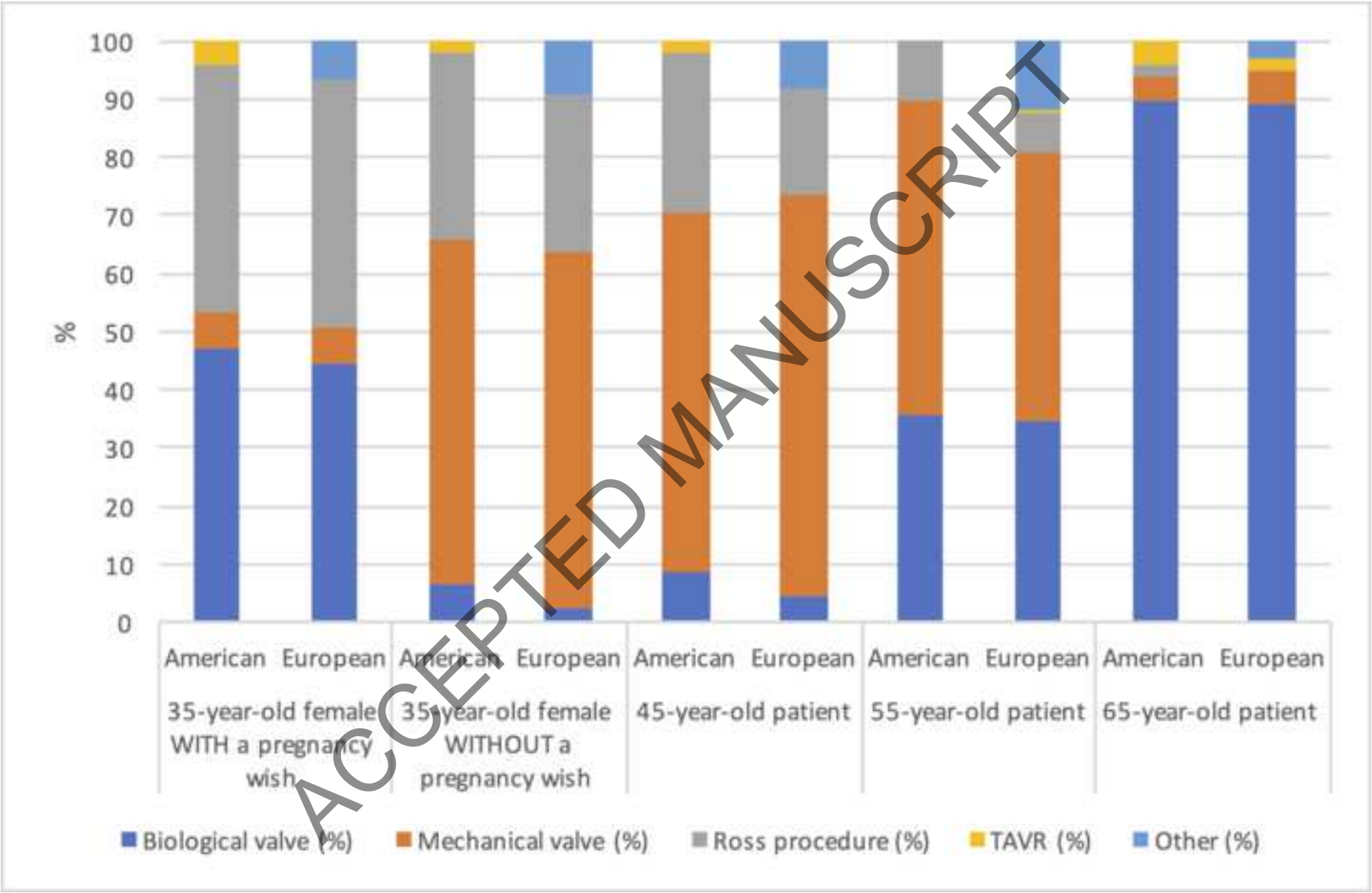

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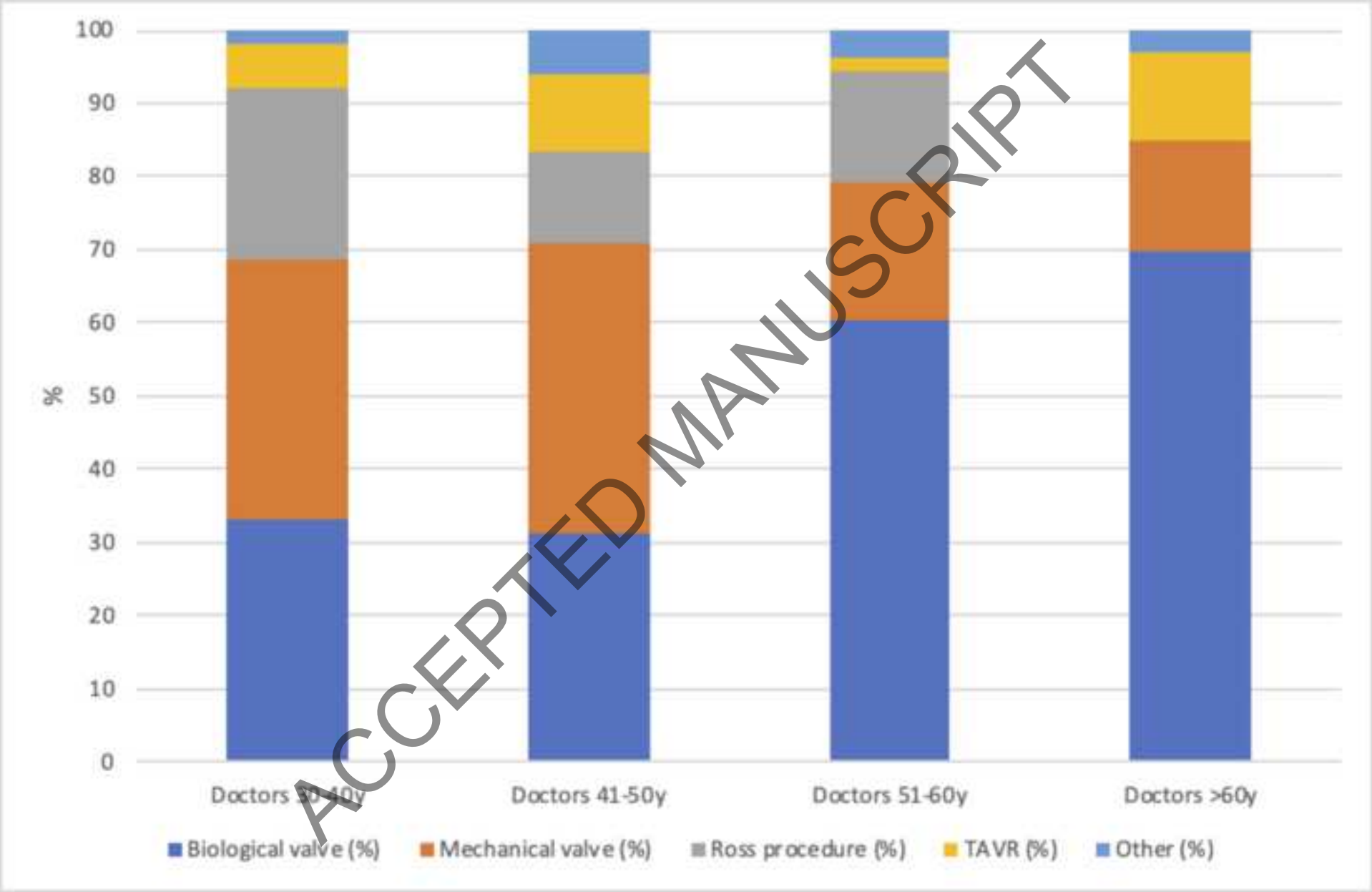

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